

Assignment: Cyber Forensic Tools	
Textbooks	
T1. Charles P. Pfleeger and Shari Lawrence Pfleeger, "Security in Computing", Pearson Education, 5th Edition, 2012 T2. Brooks, Charles J., Christopher Grow, Philip Craig, and Donald Short. Cybersecurity essentials. John Wiley & Sons, 2018 . T3. Dejeay and Murugan, "Cyber Forensics", Oxford University Press, 2018.	
References	
R1. Charles P. Pfleeger, Shari Lawrence Pfleeger, Jonathan Margulies, Security in Computing, 5th Ed, Pearson Education, 2015. R2. Behrouz A Forouzan and Debdeep Mukhopadhyay, Cryptography and Network Security, 3rd Edition, Mc Graw Hill Publication, ISBN 13: 978-93-392-2094-5. 2008.	
Web links:	
W1. https://www.youtube.com/watch?v=RYB4cG8G2xo W2. https://www.coursera.org/lecture/detecting-cyber-attacks/Cyber Security-UeDqJ , https://presiuniv.knimbus.com/user#/home	
Topics relevant to "EMPLOYABILITY SKILLS": Mobile Security for developing Employability Skills through Participative Learning techniques. This is attained through assessment component mentioned in course handout.	

Course Code: CSE319	Course Title: Machine Learning Type of Course: Theory Only	L- T-P- C	3	0	0	3
Version No.	2.0					
Course Pre-requisites	Mathematical Logic, Algebra, probability and Statistics, Vectors, Matrices.					
Anti-requisites	NIL					
Course Description	This Course aims to introduce student's concepts and techniques on Machine Learning and to study various probability based learning techniques, graphical models of Machine Learning algorithms.					

	This course encompasses various theoretical spectrum of Machine Learning concepts behind several Machine Learning algorithms without going deep into the mathematics, gaining practical experience by applying them. Covering Correlations, Regressions and to have a thorough understanding of the Supervised and Unsupervised learning techniques, and limitations on Predictive Models.			
Course Objective	The objective of the course is to familiarize the learners with the concepts of Machine Learning and attain EMPLOYABILITY SKILLS through PARTICIPATIVE LEARNING techniques			
Course Out Comes	On successful completion of the course the students shall be able to: CO 1: Explain the basic concepts on Machine Learning. [Comprehension] CO 2: Apply Supervised Machine Learning algorithms on real time Applications. [Application] CO 3: Apply Un-Supervised Machine Learning algorithm for real time problems. [Application] CO 4: Illustrate advanced concepts in machine learning [Application]			
Course Content:				
Module 1	Introduction	Assignment	Simulation/Data Analysis	6 Sessions
Introduction to Machine learning- What Why and How?, Types of Machine Learning, Applications, Models selection, Machine learning concept work flow, Issues, types of variables/features used in ML algorithms, One-hot encoding				
Module 2	Supervised learning	Assignment	Numerical from E-Resources	13 Sessions
Types of supervised learning: linear regression, Simple Linear Regression, Multiple Linear Regression, Model Evaluation, Validation and Accuracy measures for Regression models. Classification: logistic-KNN-Decision tree-SVM-Naïve Bayes, Metrics for supervised learning.				
Module 3	Unsupervised learning	Term paper/Assignment	Simulation/Data Analysis	11 Sessions
Types of Unsupervised Learning: K-means clustering, Hierarchical clustering, Association Rule Mining, Collaborative Filtering – User based and item based similarity--Applications of unsupervised learning, cluster validity measures, Components of Time Series data				
Module 4	Introduction to Neural Network	Term paper/Assignment	Simulation/Data Analysis	8 Sessions
Overview of neural networks- What and Why? , Real and artificial neurons, Threshold logic unit algorithm, Linear separability and vectors, Introduction to Learning Rules in Neural Network.				

Targeted Application & Tools that can be used: Jupyter notebook Colab notebook
Text Book Ethem Alpaydin, “Introduction to Machine Learning”, Third Edition. Stephen Marsland, “Machine Learning: An Algorithmic Perspective”, Springer, 2014, Second Edition.
References Tom M. Mitchell, “Machine Learning”, McGraw Hill Education, 2013. Sebastian Raschka and Vahid Mirjalili ,“Python Machine Learning” , PACKT Publishing, Third Edition. Wes McKinney ,“Python for Data Analysis” ,O’Reilly Media, Inc., Second Edition. Simon Haykin ,“Neural Networks: A Comprehensive Foundation”, Prentice Hall, Second Edition, 1998. Web Based Resources and E-books: W1. pu.informatics.global, https://sm-nitk.vlabs.ac.in/ W2. Udemy course on “Machine learning A-Z: Hands-on Python and R in Data Science”, https://www.udemy.com/course/machinelearning/ W3. Coursera course on “Machine learning specialization”, Andrew Ng https://www.coursera.org/specializations/machine-learning-introduction
Topics relevant to “EMPLOYABILITY SKILLS: linear regression, Classification: logistic-KNN-Decision tree-SVM-Naïve Bayes ,K-means clustering, Hierarchical clustering, Association Rule Mining for developing Employability Skills through Participative Learning techniques. This is attained through assessment component mentioned in course handout